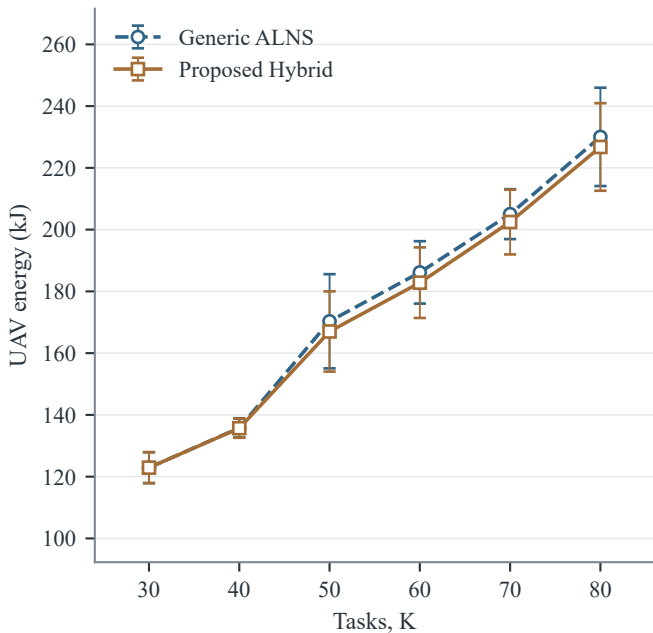
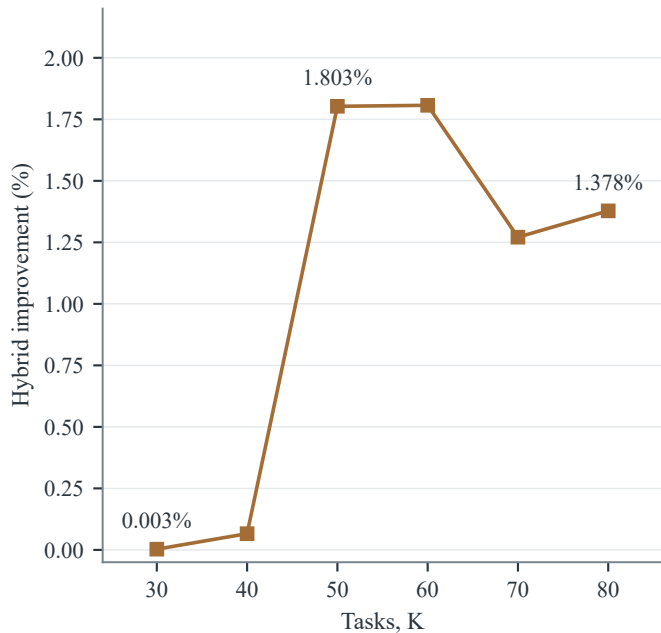
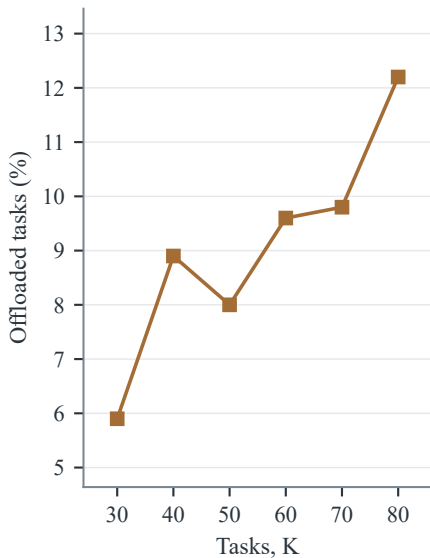
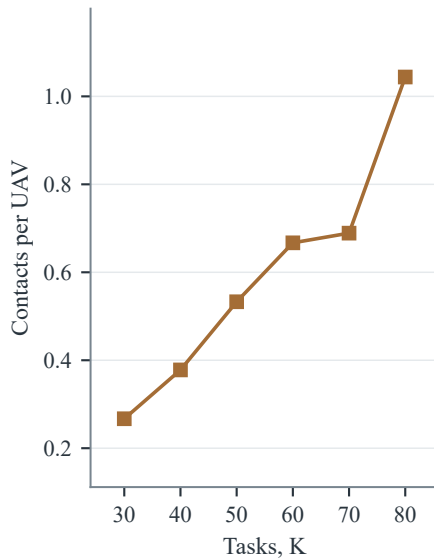
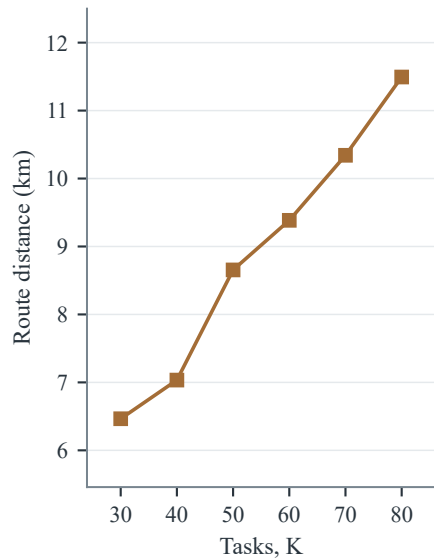


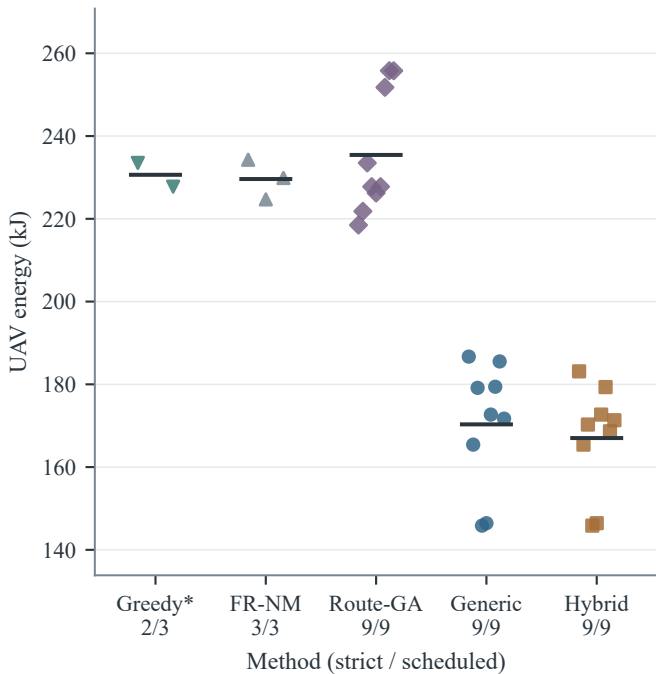
(a) Energy versus workload**(b) Paired improvement**

3 scenarios $\times$ 3 algorithm seeds per K; error bars: ± 1 SD

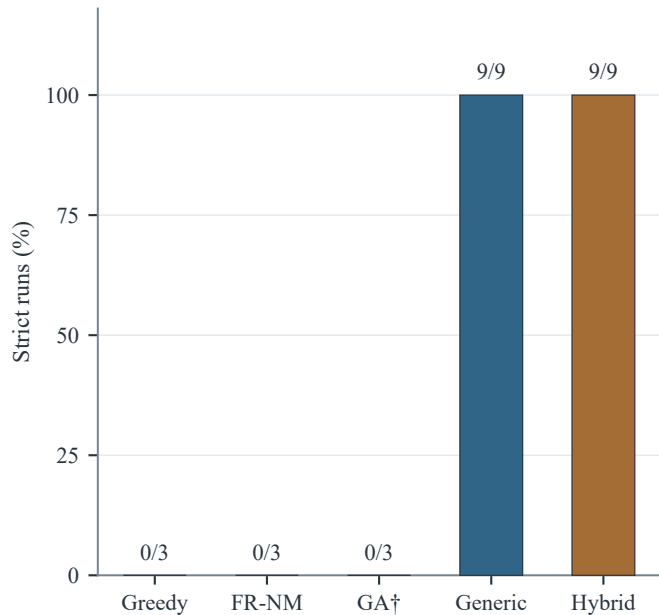
(a) Offloading**(b) MEC contacts****(c) Route length**

Proposed Hybrid; 9 Stage-1-strict runs at each task scale

(a) K = 50: run-level energy



(b) K = 80: strict feasibility



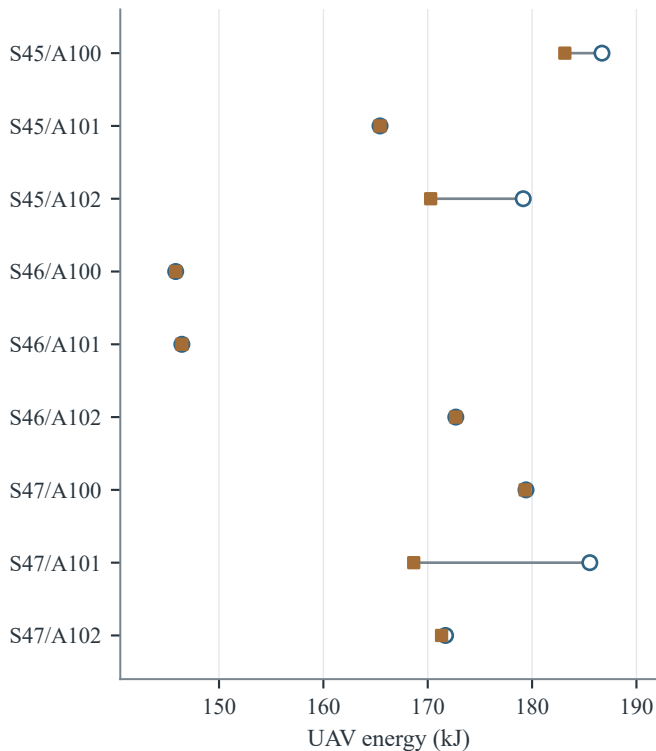
* Greedy energy uses 2 strict scenarios; horizontal marks show means.

† K=80 GA: 3-scenario pilot; search budgets are not matched.

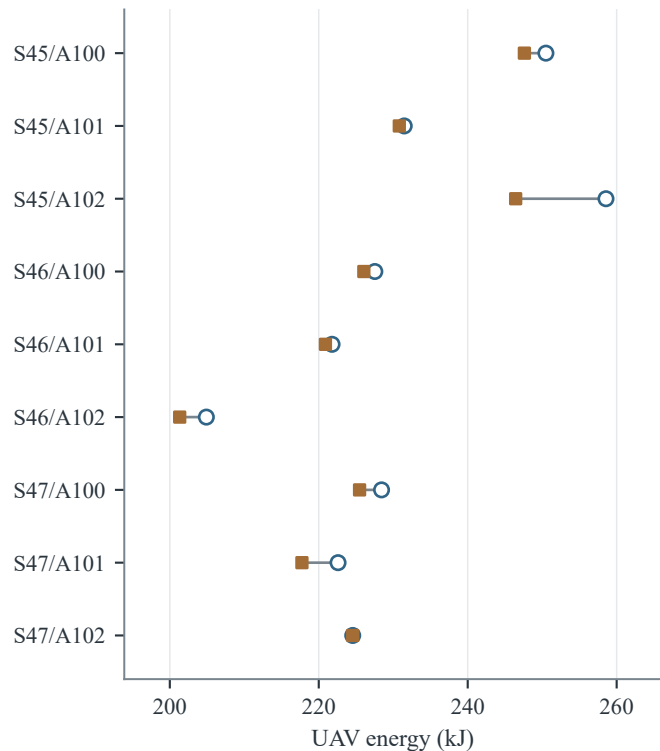
● - Generic ALNS

■ - Proposed Hybrid

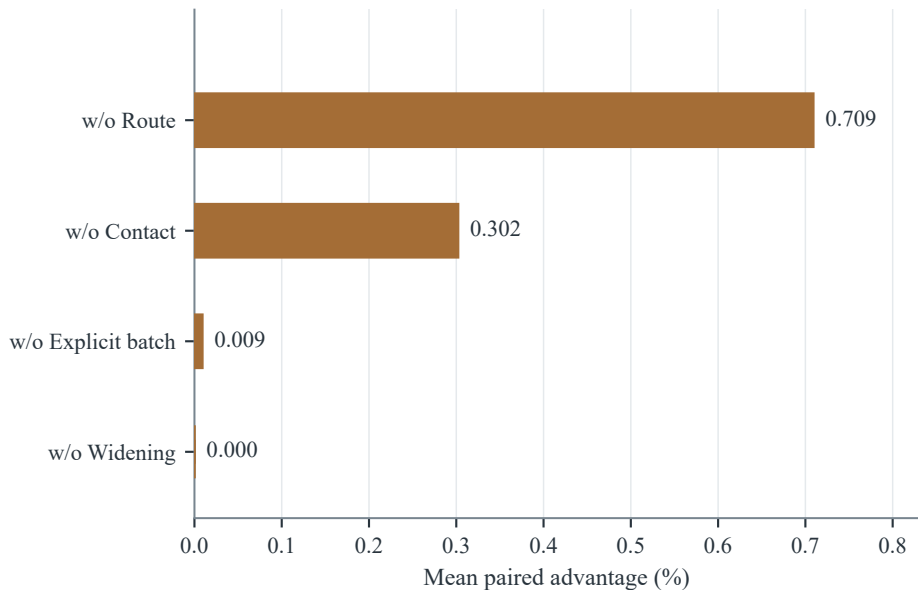
(a) K = 50



(b) K = 80



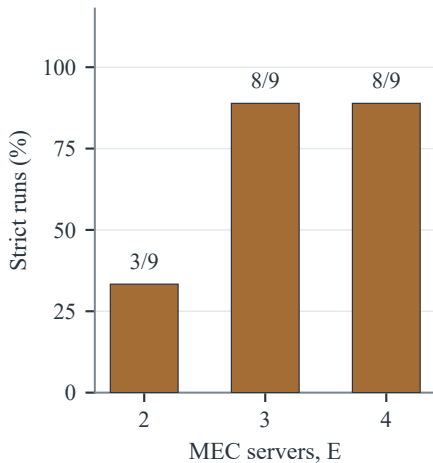
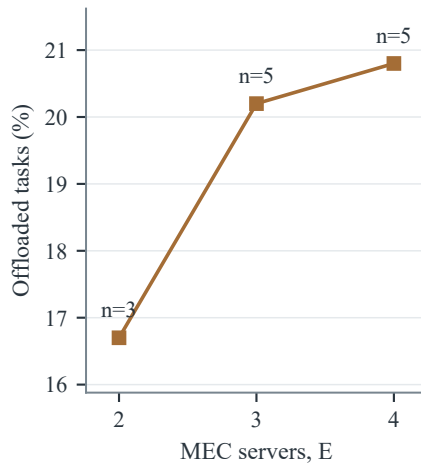
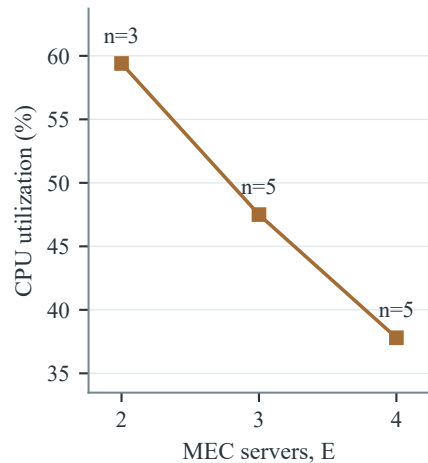
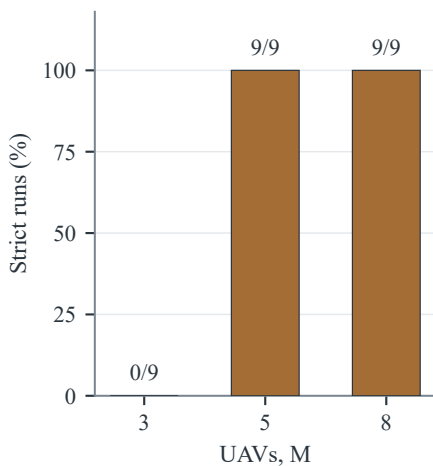
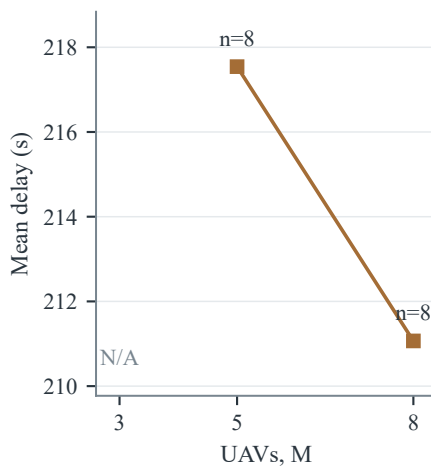
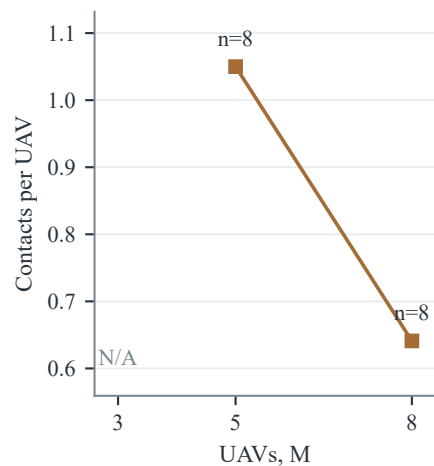
K=50: 5 better / 4 equal; K=80: 8 better / 1 equal; no worse pairs

(a) Full-Hybrid advantage

9 common-strict pairs per ablation; zero is reported explicitly

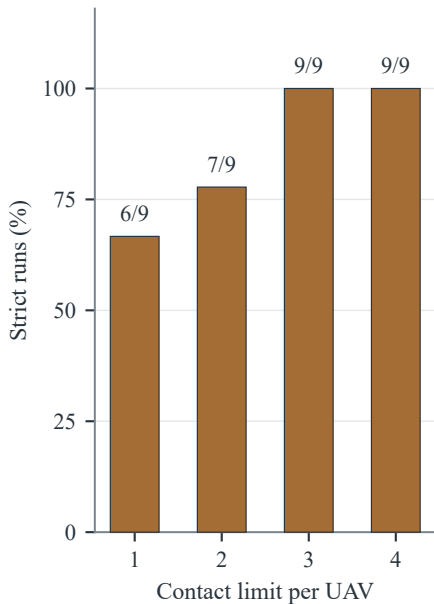
(b) Paired counts

Full better	Equal	Ablated better
6	3	0
3	5	1
2	6	1
0	9	0

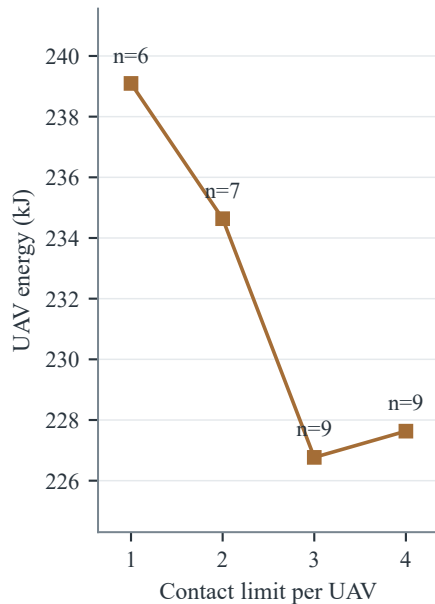
(a) MEC: strict feasibility**(b) MEC: offloading****(c) MEC: CPU occupation****(d) UAV: strict feasibility****(e) UAV: delay****(f) UAV: contacts**

Rates: Stage-1 strict / 9. Conditional metrics: Stage-2 strict n.
M=3: no strict terminal solution recovered within the search budget.

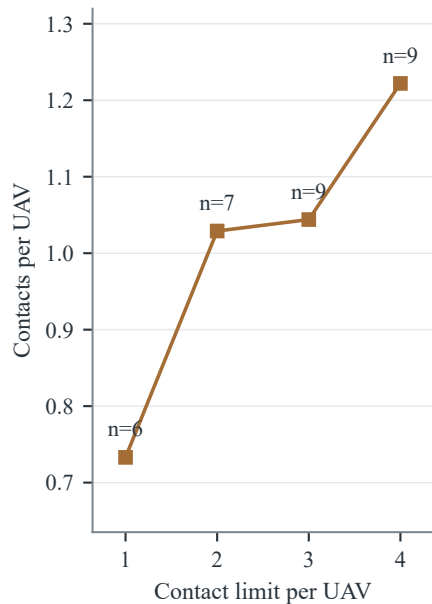
(a) Strict feasibility



(b) Conditional energy

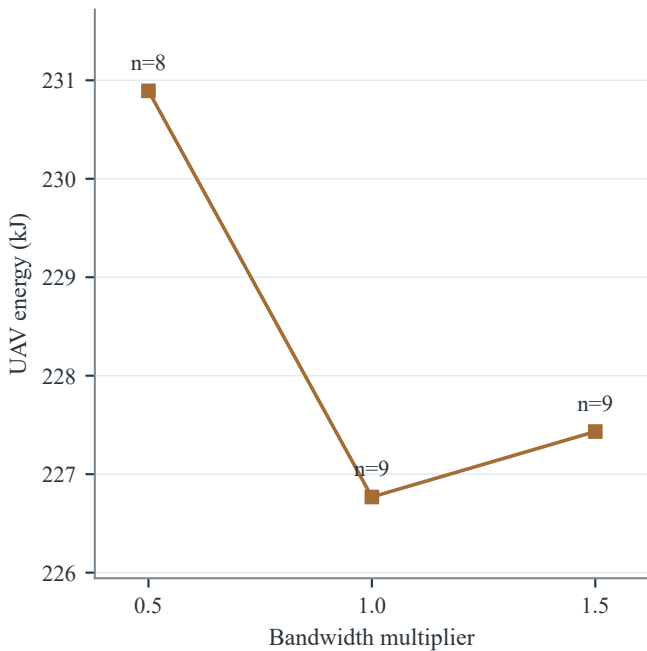


(c) Realized contacts

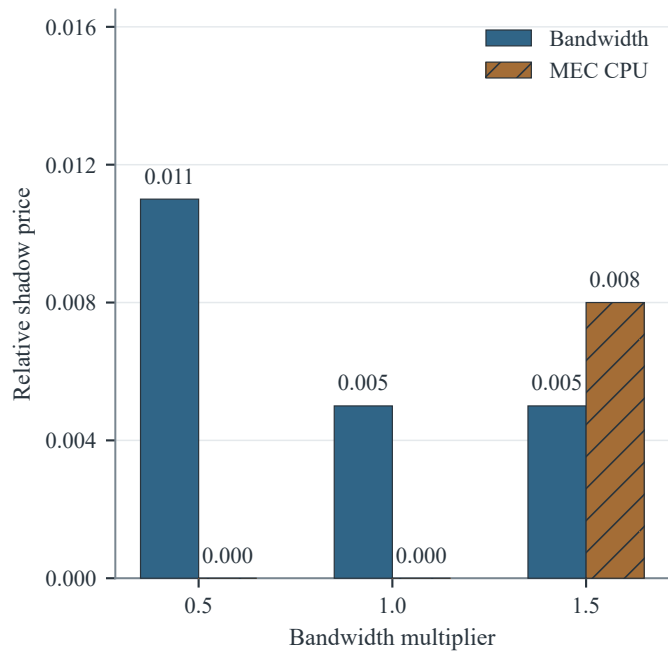


Energy and contact means use the Stage-1-strict subset shown by n.

(a) Bandwidth sensitivity



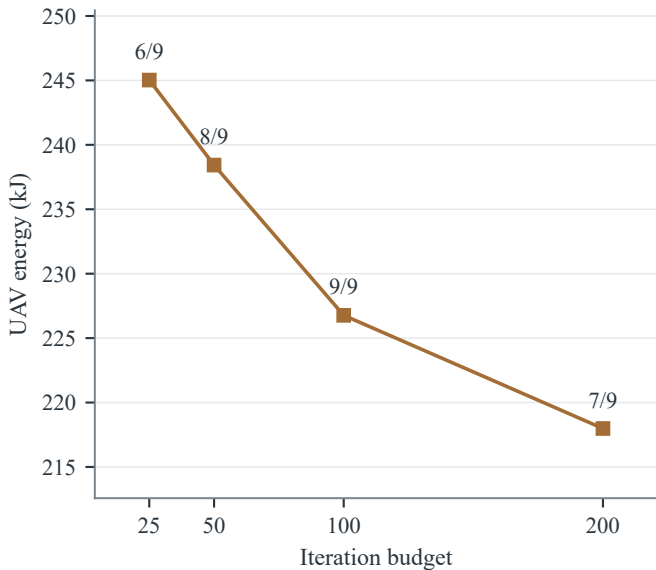
(b) Stage-1 shadow prices



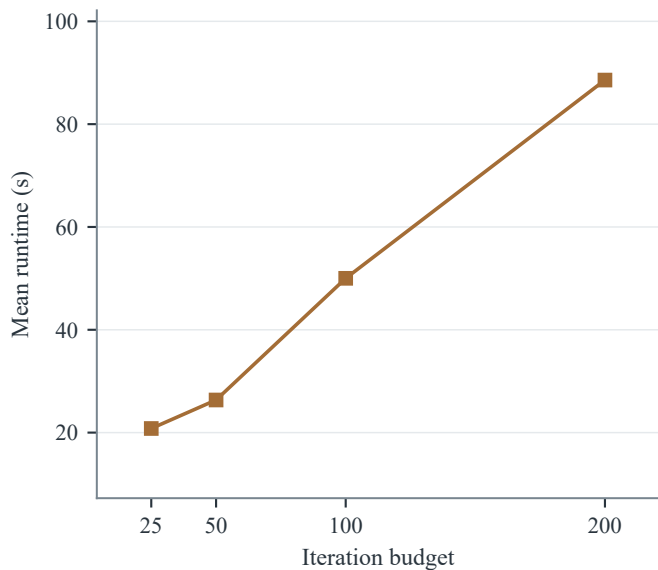
Strict subsets: 8/9, 9/9, 9/9; shadow prices use Stage-1 multipliers.

Iteration-budget sensitivity

(a) Conditional energy

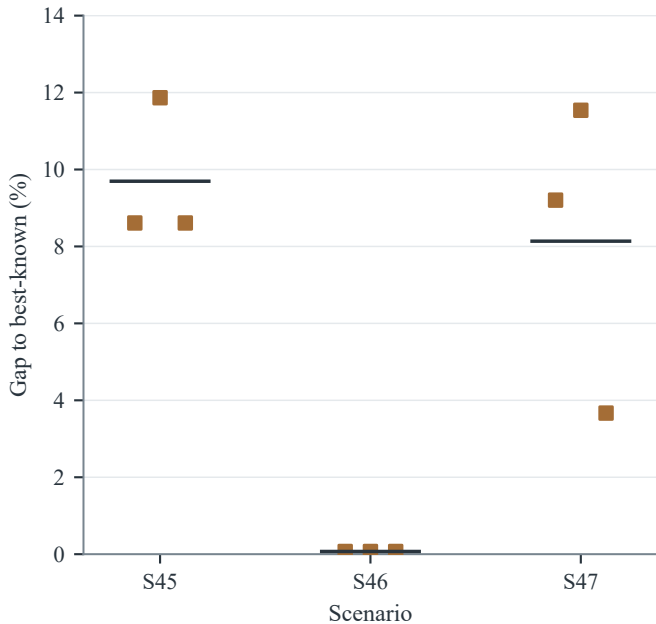


(b) Computational cost

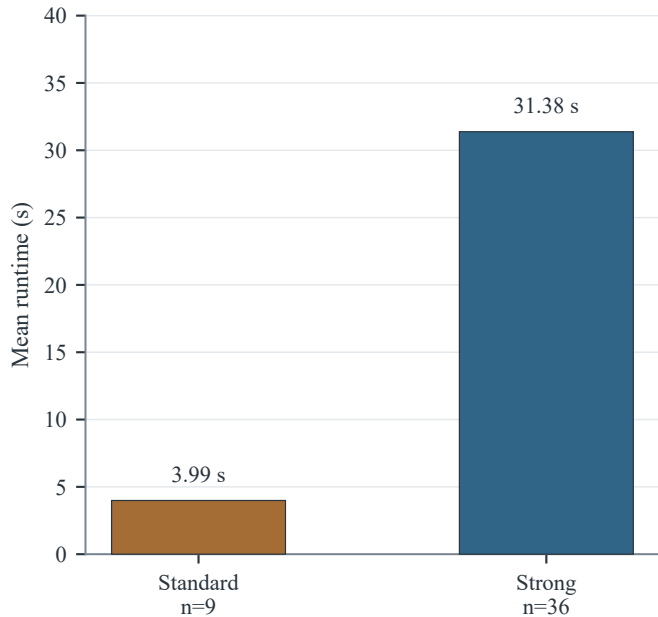


Separate budget runs; not prefixes of a single convergence trajectory.

(a) Standard-search quality

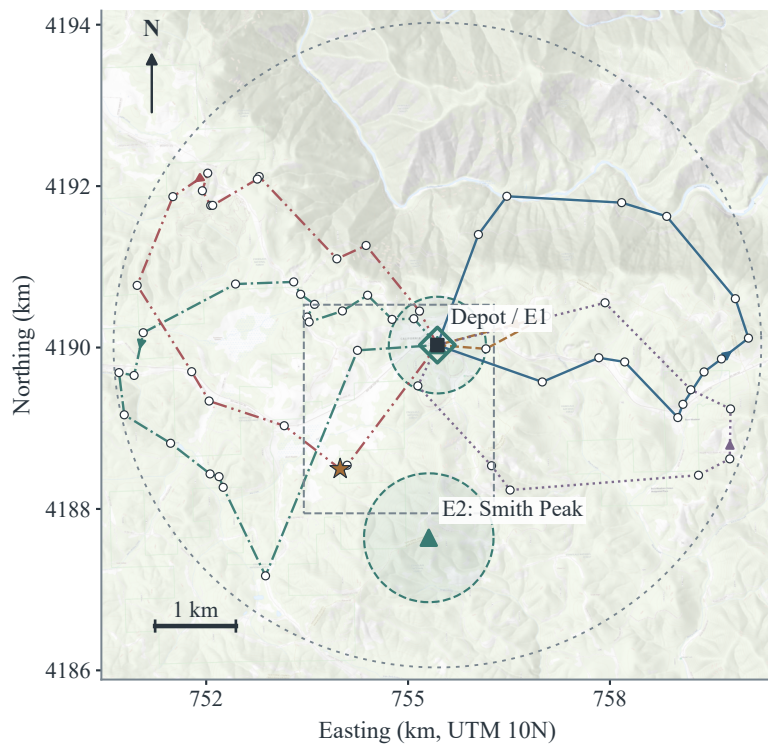


(b) Search cost

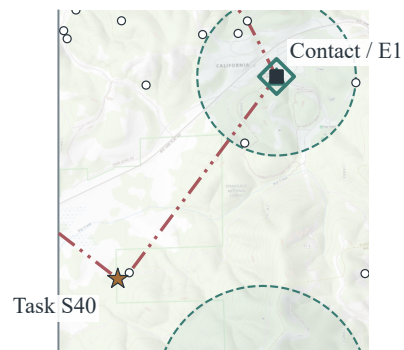


K=28, M=2, E=2; reference is empirical best-known, not a global optimum.

(a) Stanislaus: S45/A100



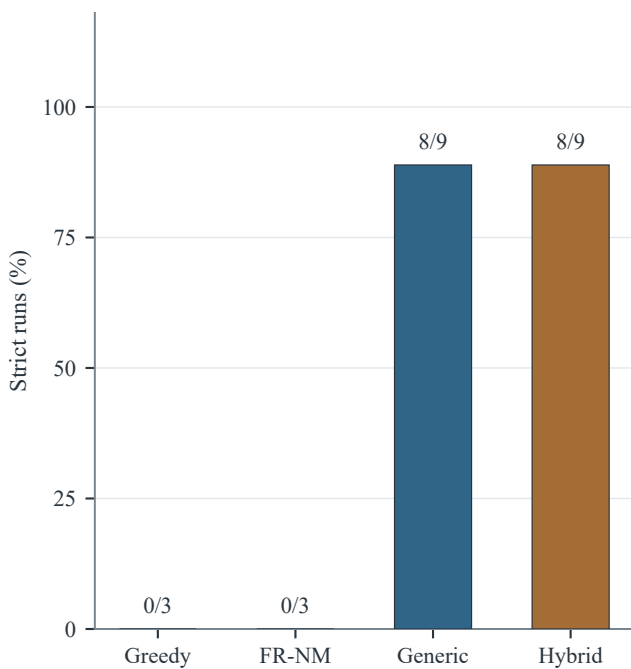
Contact detail (U5)



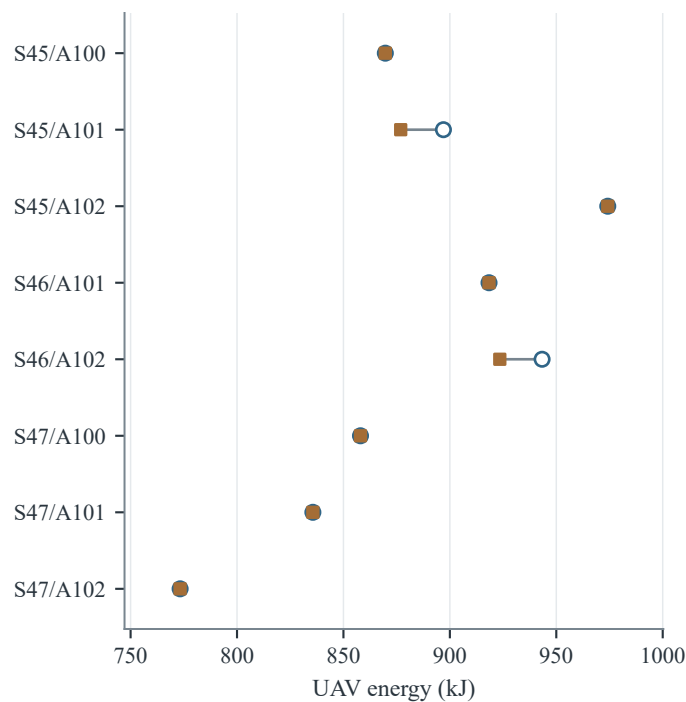
- UAV 1
- UAV 2
- UAV 3
- UAV 4
- UAV 5
- Task site
- Depot
- MEC site
- Contact
- Offloaded task
- MEC coverage
- Study area

Basemap: USGS The National Map
USFS historical fire locations
MEC servers are modeled deployments

(b) Strict feasibility

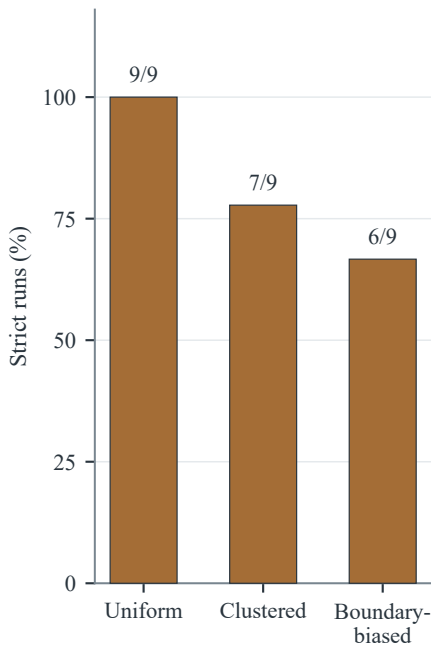


(c) Common-strict pairs

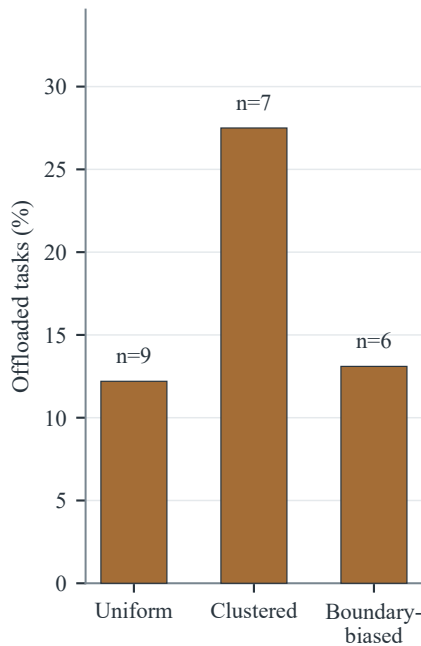


- Generic ALNS
- Proposed Hybrid

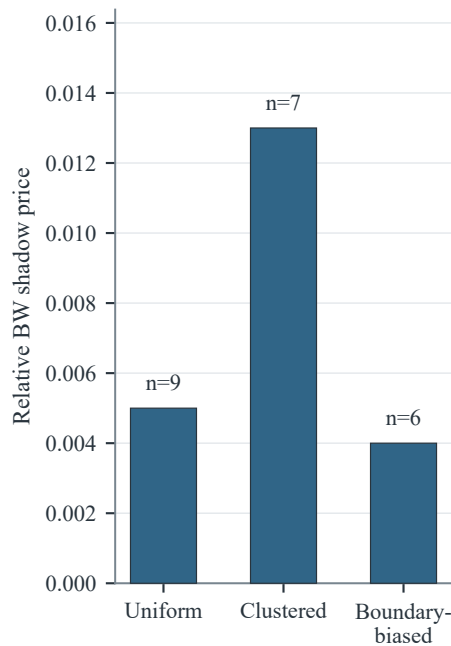
(a) Strict feasibility



(b) Offloading

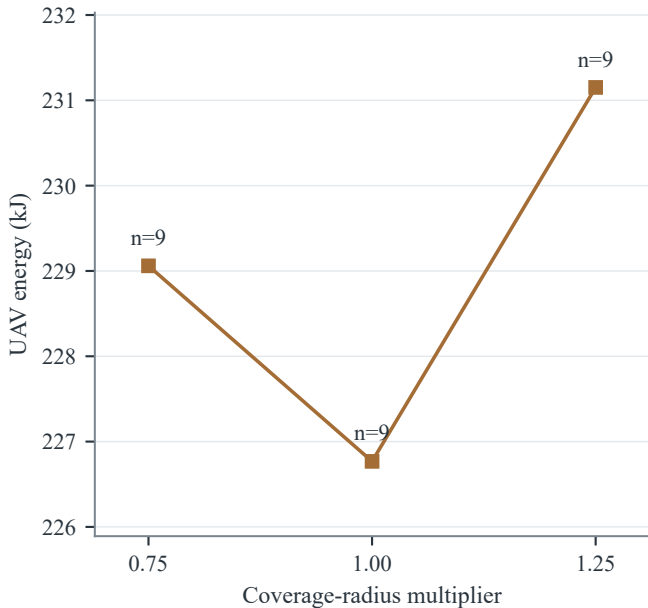


(c) Bandwidth pressure

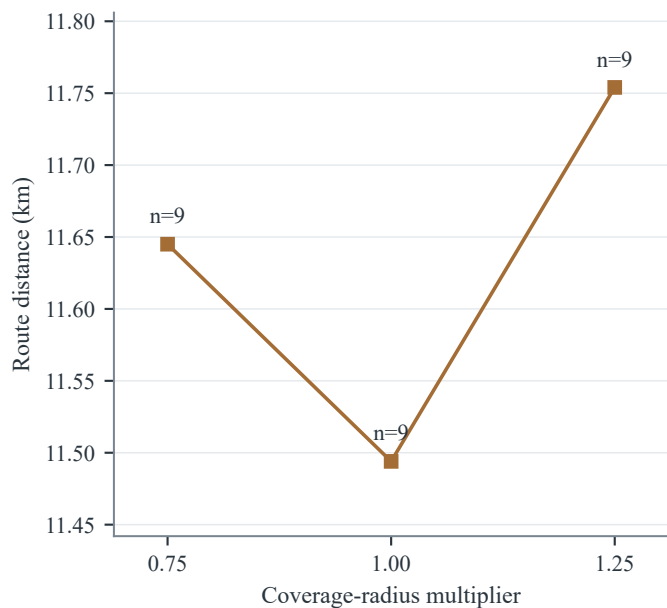


Conditional metrics use strict subsets; absolute energy is not compared.

(a) Conditional energy



(b) Route geometry



All three settings have 9/9 strict runs; larger coverage need not lower energy.